

Internship Report

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Internship: GIPEDI Internship Programme, IIT Delhi

Duration: 18/05/2026-13/07/2026

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Introduction

During this internship, I worked on a variety of embedded systems projects using the Arduino Uno, ESP32, and Seeed XIAO nRF52840 Sense development boards. The internship focused on building practical skills in embedded programming, hardware interfacing, sensor integration, communication protocols, and system design through hands-on implementation of multiple projects.

I began by developing a series of embedded applications using the Arduino Uno and ESP32, which included digital output and display interfacing, temperature monitoring and weather alert systems, a scientific calculator, a waveform generator, an RTC-based monitoring and automation system, and an audio playback system. These projects provided practical experience in interfacing different hardware peripherals, implementing control logic, and developing firmware for embedded systems. Building on the knowledge gained from these projects, I developed a Bluetooth-enabled Motion Data Logger using the **Seeed XIAO nRF52840 Sense**. The system acquires motion data from the onboard IMU sensor, records timestamped data on a microSD card using a Real-Time Clock (RTC), communicates with an Android application over Bluetooth Low Energy (BLE), and incorporates deep sleep functionality for low-power operation. As part of this project, I also developed the companion Android application, designed a custom PCB for the hardware, and prepared comprehensive technical documentation.

This report presents the hardware platforms, software tools, technologies, and libraries used during the internship, along with the projects completed, challenges encountered, and the practical knowledge and skills acquired throughout the internship.

Hardware and Software Resources

Development Boards

- Arduino Uno
- Seeed XIAO nRF52840 Sense
- Espressif ESP32-WROOM 32

Sensors

- LSM6DS3 IMU(XIAO inbuilt)
- LM35

Communication Modules

- Bluetooth Low Energy (BLE)
- USB (microUSB, type-C)

Storage Devices

- microSD Card Module

Displays

- Nokia PCD8544 LCD
- 16×2 Character LCD(JHD 162A)
- LT543 Hi-lite(7 segment display)

Other Components

- Push Buttons
- LEDs
- Buzzer
- Relay Module
- L293 Motor Driver
- Potentiometers
- DS1302 Real-Time Clock
- DHT22 Temperature Sensor
- R-2R DAC
- Breadboard
- Resistors
- Connecting Wires

Development Tools and Libraries

This chapter describes the software tools, development environments, frameworks, board support packages, and software libraries used throughout the internship for firmware development, mobile application development, PCB design, documentation, and version control.

Development Software

Software	Version	Purpose
Arduino IDE	Version: 2.3.10 Date: 2026-06-09T16:44:44.168Z CLI Version: 1.5.1 Copyright © 2026 Arduino s.r.l. and/or its affiliated companies	Embedded firmware development
Visual Studio Code	Version: 1.127.0 (Universal) Commit: 4fe60c8b1cdac1c4c174f2fb180d0d758272d713 Date: 2026-06-30T10:52:33+02:00 Electron: 42.2.0 ElectronBuildId: 14159160 Chromium: 148.0.7778.97 Node.js: 24.15.0 V8: 14.8.178.14-electron.0 OS: Darwin arm64 25.5.0	Source code editing
Flutter SDK (Dart)	Flutter 3.44.4 • channel stable • https://github.com/flutter/flutter.git Framework • revision ad70ec4617 (13 days ago) • 2026-06-24 11:07:06 -0700 Engine • hash 700aebeca4c0e610f109a3979ee3e71b69d666bc (revision a10d8ac38d) (13 days ago) • 2026-06-23 23:09:55.000Z Tools • Dart 3.12.2 • DevTools 2.57.0	Android application development
Android Studio	Quail 1 2026.1.1 Patch	Android build and debugging
Obsidian	Version 1.12.7 (Installer 1.11.7)	Technical documentation
KiCad	Version: 10.0.3, release build Libraries: wxWidgets 3.2.8 FreeType 2.14.3 HarfBuzz 12.3.2 FontConfig 2.17.1 libcurl/8.7.1 (SecureTransport) LibreSSL/3.3.6 zlib/1.2.12 nghttp2/1.68.1	PCB design

Software	Version	Purpose
	Platform: macOS Version 26.5.2 (Build 25F84), 64 bit, Little endian, wxMac Build Info: Date: May 13 2026 20:28:11 wxWidgets: 3.2.8 (wchar_t,wx containers) Boost: 1.90.0 OCC: 7.9.3 Curl: 8.7.1 ngspice: 44.2 Compiler: Clang 17.0.0 with C++ ABI 1002 KICAD_IPC_API=ON KICAD_USE_PCH=OFF	
GitHub	2.54.0	Source code management
SimulIDE	1.1.0 SR2	Simulation of Circuits
LTSpice		Wiring Check
nRF connect		Bluetooth connection to smartphone

Documentation and Design Tools

Tool	Purpose
Markdown	Technical documentation
Mermaid.js	Flowchart and system diagram generation
Obsidian	Project documentation and note management
Git	Documentation version control

Arduino Board Support Packages

Package	Purpose
Arduino AVR Boards	Arduino Uno development
ESP32 by Espressif Systems	ESP32 firmware development
Seeed nRF52 Boards	Seeed XIAO nRF52840 Sense support

Arduino Libraries

Core Libraries

Library	Purpose
Wire	I ² C communication
SPI	SPI communication
SdFat	microSD card interface
EEPROM	Non-volatile data storage

Sensor Libraries

Library	Purpose
LSM6DS3	IMU sensor communication
DHT	DHT22 temperature and humidity sensor
OneWire	OneWire communication protocol
DallasTemperature	DS18B20 temperature sensor

Communication Libraries

Library	Purpose
Adafruit Bluefruit	Bluetooth Low Energy (BLE) communication

RTC Libraries

Library	Purpose
RtcDS3231 (<i>or RTCLib</i>)	Real-Time Clock communication

Display Libraries

Library	Purpose
LiquidCrystal	16×2 LCD display
Adafruit_GFX	Graphics rendering
Adafruit_PCD8544	Nokia 5110 LCD display

Input Libraries

Library	Purpose
Keypad	Matrix keypad interface

Flutter Packages

Package	Version	Purpose
flutter_blue_plus	2.3.10	Bluetooth Low Energy communication
permission_handler	12.0.3	Runtime permission management
file_picker	10.3.10	File selection from device storage
shared_preferences	2.5.3	Local application settings
url_launcher	6.3.1	Launch external URLs and file locations
path_provider	2.1.5	Access application storage directories
cupertino_icons	1.0.8	iOS-style icon set

Development Utilities

Utility	Purpose
Arduino Serial Monitor	Firmware debugging
GitHub	Source code repository
USB Serial Communication	Data monitoring and debugging
BLE Scanner / Android Application	Bluetooth testing and validation

Programming Languages

Language	Application
C++	Embedded firmware development
Dart	Android application development
Markdown	Technical documentation
YAML	Flutter project configuration

Bill of Materials (BOM)

The table below lists the major hardware components used in the development of the Motion Data Logger.

Sl. No.	Component	Quantity	Purpose
1	Seeed XIAO nRF52840 Sense	1	Main microcontroller with onboard IMU
2	DS3231 Real-Time Clock (RTC) Module	1	Accurate timestamp generation
3	microSD Card Module	1	Storage interface for logged data
4	microSD Card	1	Motion data storage
5	Li-Po Battery (3.7 V)	1	Portable power supply
6	Battery Charging/Power Management Circuit	1	Battery charging and power management
7	Push Button	1	User input for logging, BLE mode, and deep sleep
8	Custom Printed Circuit Board (PCB)	1	Hardware integration
9	USB Type-C Connector	1	Programming and battery charging
10	Pin Headers / Connectors	As required	External electrical connections
11	Decoupling Capacitors	As required	Power supply filtering and stability
12	Pull-up/Pull-down Resistors	As required	Signal conditioning and input stabilization

Knowledge and Technologies Acquired

During the internship, I learned and applied several concepts related to embedded systems and software development.

The major areas covered include:

- Embedded C/C++ Programming
 - Arduino Development
 - GPIO Programming
 - Analog-to-Digital Conversion (ADC)
 - PWM Signal Generation
 - Digital Input/Output Interfacing
 - Sensor Interfacing
 - I²C Communication
 - SPI Communication
 - UART Communication
 - Bluetooth Low Energy (BLE)
 - SD Card File Handling
 - Real-Time Clock Integration
 - Power Management
 - Deep Sleep Implementation
 - Android Application Development using Flutter
 - PCB Design using KiCad
 - Technical Documentation using Markdown
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Work Completed During the Internship

This chapter summarizes the practical work completed throughout the internship.

1. Simluation and Practical Application of Embedded circuits

The following embedded system applications were designed and implemented during the initial phase of the internship.

- Digital Output and LED,DISPLAY Interfacing
- Temperature Monitoring and Weather Monitoring System
- Scientific Calculator

- Waveform Generator
- RTC-Based Monitoring System
- Audio Playback System

Each task focused on understanding different hardware interfaces, communication protocols, peripheral integration, and embedded programming techniques.

2. Project of Data Recorder

The primary project undertaken during the internship was the development of a Bluetooth-enabled Motion Data Logger.

The project involved designing an embedded system capable of acquiring motion data, storing measurements on a microSD card, communicating with a mobile application over Bluetooth Low Energy, and operating efficiently using deep sleep power management.

Major components of the project include:

- System Architecture
- Hardware Design
- Firmware Development
- IMU Integration
- RTC Integration
- Data Logging
- Bluetooth Communication
- Android Application Development
- Deep Sleep Implementation
- PCB Design
- System Testing and Validation

Challenges Encountered

Throughout the internship, several technical challenges were encountered during both hardware and software development.

Some of the major challenges included:

- Bluetooth communication reliability
- SD card data logging
- RTC synchronization
- Power management
- Deep sleep wake-up implementation

- File transfer over BLE
- PCB routing
- Hardware debugging
- Firmware debugging
- Mobile application debugging

Addressing these challenges significantly improved my understanding of embedded system development and debugging methodologies.

Skills Developed

Technical Skills

- Embedded Systems Development
 - Arduino Programming
 - Embedded C/C++
 - Bluetooth Low Energy
 - Sensor Interfacing
 - PCB Design
 - Android Application Development
 - Version Control using Git
 - Technical Documentation
 - Debugging
 - Problem Solving
 - Technical Writing
 - Project Planning
 - Hardware Integration
 - Firmware Development
 - Software Development
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Conclusion

The internship provided valuable practical experience in embedded systems, firmware development, hardware interfacing, wireless communication, mobile application development, and PCB design.

Through the completion of multiple laboratory tasks and the development of the Motion Data Logger project, I gained hands-on experience in designing complete embedded solutions while improving my

debugging, documentation, and software development skills.

The knowledge acquired during this internship has strengthened my understanding of embedded system design and provided a strong foundation for future academic and research work.

References

- [Arduino Documentation](#)
- [Seeed Studio Documentation](#)
- [Flutter](#)
- [KiCad Tutorial](#)
- Component Datasheets([DS1302](#), [DS3231](#), [XIAO nRF52840 Sense](#), [ESP32](#), [Display](#))
- AI([ChatGPT](#), [Copilot](#))